

BUSINESS REQUIREMENTS DOCUMENT

Inventory Management Agent System

AI-Powered Multi-Agent System for Retail Operations

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Document Control

Document Title	Inventory Management Agent - Business Requirements Document
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Executive Summary

This document defines the business requirements for an AI-powered Inventory Management Agent System designed for retail operations. The system leverages a multi-agent architecture deployed on Microsoft Azure to automate and optimize critical inventory management processes across warehouses and retail stores.

Business Problem

RetailCo currently faces significant operational challenges in inventory management, including manual stock monitoring, reactive replenishment decisions, inefficient vendor selection, and lack of real-time visibility across 250+ locations. These issues result in stockouts costing approximately \$12M annually, excess inventory holding costs of \$8M, and missed revenue opportunities estimated at \$15M.

Proposed Solution

The Inventory Management Agent System implements an agentic AI architecture with specialized agents handling distinct responsibilities: inventory monitoring, demand forecasting, replenishment planning, transfer optimization, supplier evaluation, policy enforcement, procurement execution, and notifications. The system will integrate with existing ERP, POS, and warehouse management systems to provide autonomous decision-making capabilities while maintaining appropriate governance controls.

Business Context

Current State Analysis

Organizational Profile

RetailCo operates a national retail chain with 250 stores across 35 states, 8 regional distribution centers, and annual revenue of \$2.4B. The company maintains an average inventory of 45,000 active SKUs with seasonal variations reaching 62,000 SKUs during peak periods. Current inventory turnover rate is 6.2x annually, below the industry benchmark of 8.5x.

Current Process Challenges

Process Area	Current Challenge	Business Impact
Stock Monitoring	Manual daily reviews, 24-48 hour lag in identifying low stock situations	Average 180 stockout events per week, 8.5% lost sales rate
Demand Forecasting	Basic historical averaging, no ML capabilities, limited seasonality handling	68% forecast accuracy, 32% excess inventory during off-peak periods
Replenishment	Fixed reorder points, no dynamic adjustment, manual purchase order creation	12-hour average decision time, suboptimal order quantities, frequent rush orders
Vendor Selection	Spreadsheet-based comparison, incomplete cost analysis, inconsistent criteria	15% higher procurement costs, missed volume discount opportunities

System Overview

Multi-Agent Architecture

The Inventory Management Agent System implements a distributed agentic architecture where specialized AI agents collaborate to manage the complete inventory lifecycle. Each agent operates autonomously within defined governance boundaries while coordinating with other agents through an orchestration layer.

Agent Taxonomy

Group	Agent Name	Core Functions	Value Proposition
Control	Orchestrator Handlers	Workflow coordination, agent routing, state management	Ensures agent collaboration and prevents conflicting decisions
	Registry Repository Service	Agent discovery, capability lookup, metadata management	Decouples agent implementation from orchestration logic
	Workflow Repository Service	State persistence, workflow history, replay capabilities	Enables audit trails and recovery from failures
Domain	Inventory Monitoring Agent	Real-time stock tracking, shortage detection, threshold monitoring	Proactive identification of inventory issues before stockouts occur
Domain	Demand Forecast Agent	ML-based forecasting, seasonality analysis, confidence scoring	Improves forecast accuracy from 68% to target 88%
Domain	Replenishment Planning Agent	Order quantity optimization, urgency assessment, coverage gap analysis	Reduces decision time from 12 hours to under 5 minutes
Domain	Transfer Optimization Agent	Surplus detection, inter-location transfer recommendations, cost optimization	Reduces unnecessary procurement by utilizing existing inventory

Domain	Supplier Evaluation Agent	Multi-criteria vendor comparison, total cost analysis, performance scoring	Achieves 15% cost reduction through optimal vendor selection
Governance	Policy Approval Agent	Threshold enforcement, rule validation, approval workflow routing	Ensures compliance while enabling automated execution within limits
Execution	Procurement Agent	Purchase order creation, ERP integration, document generation	Eliminates manual PO entry, reducing processing time by 90%
Execution	Notification Agent	Alert distribution, stakeholder notification, exception reporting	Keeps stakeholders informed without manual status updates

Functional Requirements

FR-1: Real-Time Inventory Monitoring

Requirement ID	FR-1
Description	The system shall monitor stock levels across all warehouses and stores in real-time, tracking inventory positions, detecting low stock situations, and identifying potential overstock conditions.
Business Value	Reduces stockout incidents
Priority	High - Critical path dependency

FR-2: Demand Forecasting

Requirement ID	FR-2
Description	The system shall generate demand forecasts for each SKU by location using machine learning algorithms that incorporate historical sales data, seasonality patterns, promotional calendars, and external factors.
Business Value	Improves forecast accuracy
Priority	High - Critical path dependency

FR-3: Automated Replenishment Planning

Requirement ID	FR-3
Description	The system shall automatically calculate recommended replenishment quantities based on demand forecasts, current inventory positions, supplier lead times, and defined service level targets.
Business Value	Reduces replenishment decision time
Priority	High
Acceptance Criteria	• Order quantities optimize total cost including purchase, holding, and shortage costs • Urgency levels assigned (critical, high, medium, low) based on stockout risk • Coverage gap analysis showing days until stockout at current demand rate • Supplier lead times and minimum order quantities respected

FR-4: Intelligent Vendor Selection

Requirement ID	FR-4
Description	The system shall evaluate and compare vendor options based on multiple criteria including unit cost, lead time, historical performance, approval status, and payment terms to recommend optimal suppliers for each procurement decision.
Business Value	procurement cost reduction through optimal vendor selection and volume discount utilization
Priority	Medium
Acceptance Criteria	• Total cost calculation includes unit price, shipping, duties, and payment term value • Vendor performance scoring based on on-time delivery, quality, and historical fill rate • Approved vendor list enforced per compliance requirements • Lead time comparison enables fastest delivery when urgency is critical

FR-5: Policy Enforcement and Governance

Requirement ID	FR-5
Description	The system shall enforce business rules, approval thresholds, preferred vendor requirements, and human-in-the-loop conditions to ensure compliance with organizational policies while enabling automation within approved boundaries.
Business Value	Balances automation benefits with governance requirements, maintains audit compliance
Priority	High - Regulatory requirement

Non-Functional Requirements

Performance Requirements

Metric	Target	Measurement
Inventory data refresh latency	< 15 minutes	95th percentile end-to-end
Forecast generation time	< 30 seconds per SKU	Median response time
Replenishment decision latency	< 5 minutes	From alert to recommendation
System availability	99.9% uptime	Monthly measurement

Security and Compliance

- Role-based access control (RBAC) enforced for all agent operations and human approvals
- Data encryption at rest using Azure Storage Service Encryption and in transit using TLS 1.3
- Comprehensive audit logging of all agent decisions, approvals, and system modifications
- SOC 2 Type II compliance for data handling and processing
- Automated backup and disaster recovery with 4-hour RPO and 8-hour RTO targets

Scalability

- Support for 100,000+ active SKUs across 500 locations without performance degradation
- Horizontal scaling capability to handle 3x peak seasonal load increases
- Agent-based architecture enables independent scaling of individual capabilities

Appendix A: Synthetic Test Data Specifications

For POC development and testing, the following synthetic datasets will be generated to simulate RetailCo operations:

Store and Warehouse Locations

- 50 retail store locations across 10 states (California, Texas, Florida, New York, Illinois, Ohio, Georgia, North Carolina, Michigan, Pennsylvania)
- 3 regional distribution centers (West Coast, Central, East Coast)
- Store formats: Small (5,000 sq ft), Medium (12,000 sq ft), Large (25,000 sq ft)

Product Catalog

- 5,000 SKUs across 8 product categories: Electronics, Apparel, Home Goods, Sporting Goods, Toys, Health & Beauty, Grocery, Seasonal
- Price range: \$4.99 to \$1,299.99
- Velocity distribution: 20% fast-moving (A items), 30% medium (B items), 50% slow-moving (C items)
- Seasonal patterns: Holiday peaks (Nov-Dec), Back-to-School (Aug-Sep), Summer (Jun-Jul)

Historical Sales Transactions

- 24 months of daily transaction data per store location
- Average 450 transactions per store per day with realistic variance
- Day-of-week patterns: Weekend sales 30% higher than weekdays
- Promotional events: 15% of weeks include promotion with 25-40% sales lift

Inventory Positions

- Current stock levels for all SKU-location combinations
- Stock distribution: 15% below safety stock, 60% optimal range, 20% overstock, 5% zero stock
- In-transit inventory: 8% of total system inventory in transit at any time

Supplier Database

- 35 vendor profiles with varying characteristics
- Lead time range: 2-21 days depending on vendor and product category
- Pricing tiers: Volume discounts at 100, 500, 1000+ unit thresholds
- Performance metrics: On-time delivery rate (85-99%), quality score (90-100%), fill rate (92-100%)

- Payment terms: Net 30, Net 60, 2/10 Net 30 variations

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